

PHY 111 – GENERAL PHYSICS I

1. SPACE, TIME & MOTION {Units & Dimensions}

Space: Space is defined as a boundless 3-dimensional extent in which objects and events occur with relative position and direction. Each object in the universe exist at a particular point in space and time. Also, an object in motion undergoes a continuous change in its position within the stationary 3-dimensional matrix space.

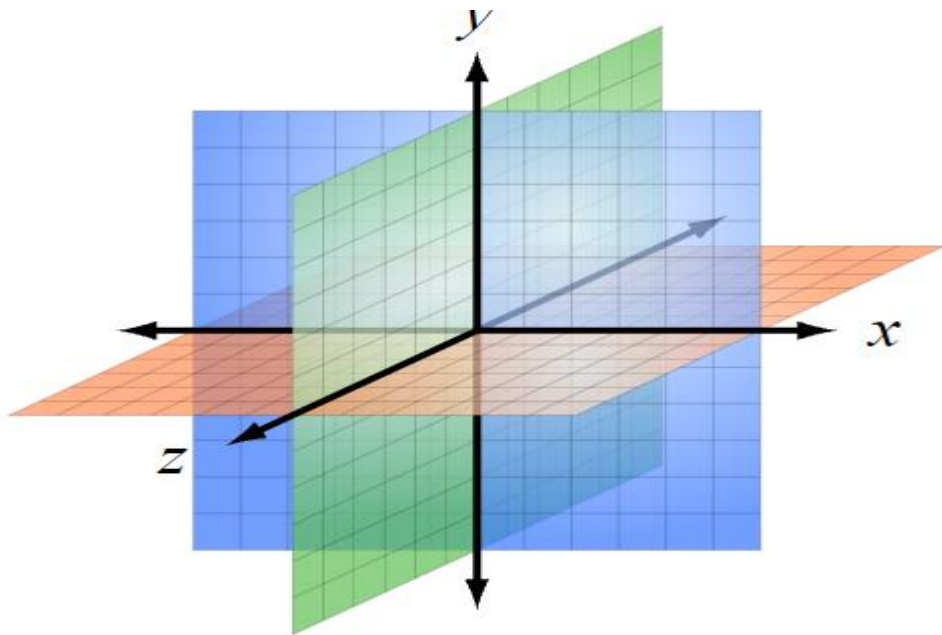


Fig. 1. 3-dimensional matrix space

According to Isaac Newton's view, space is absolute i.e. it exist independently and permanently, whether there is any matter in the space or moving through it. Absolute space is its own nature without depending on anything external and always remains similar. Immoveable space is thus, a sort of stationary 3-dimensional matrix in which one can place objects or through which objects can move without producing any interaction between the objects and space.

Time: Time simply means duration which is defined as a part of the measuring system used to sequence events, compare the duration of events and also the intervals between them. Time is a fundamental physical quantity which is used to define other quantities e.g. speed, velocity, acceleration, deceleration etc.

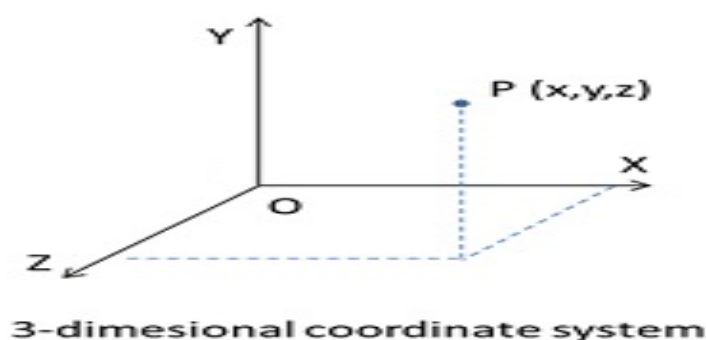
According to Isaac Newton's view, time is also absolute because it flows on without regards to any physical object or event. i.e. one can neither speed up time nor slow down its rate and thus, the flow of time exists uniformly throughout the universe.

Motion: Motion is defined as the change in position of an object with respect to time. Motion is typically described in terms of distance, displacement, time, speed, velocity, acceleration. A body which does not move is said to be at rest, motionless, stationary, immobile or constant/fixed position.

Motion is always observed and measured relative to a frame of reference. Since, there is no absolute frame of reference, absolute motion cannot be determined; this is emphasized by the term "relative motion".

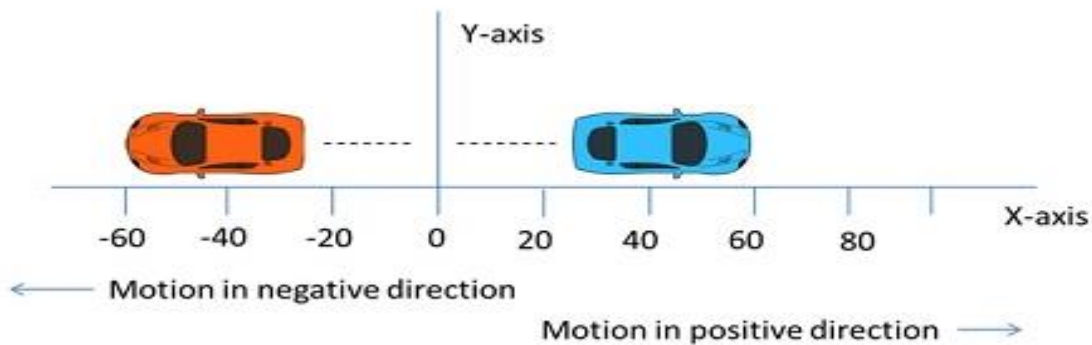
A **frame of reference** may refer to a coordinate system or set of axes (x, y, z) within which to measure the position, orientation and other properties of objects in it.

To specify the position of a particle in a space, choose 3 mutually intersecting lines called the reference lines;



Examples

1. Motion along a fixed straight line: Consider a particle undergoing motion along a fixed straight line, the position of the particles along the straight line is determined by choosing any point along the straight line called **reference point**.



Since, the particle in motion can be on either side of the reference point, the motion in one direction is considered positive and the motion on the other direction is considered negative.

2. Motion on a plane: there are two methods of specifying the position of a particle in a plane namely are:
 - i. Choose any 2 perpendicular lines which is called the reference lines and their point of intersection is called the origin of rectangular coordinates. The description of the motion of the particle P can be done in terms of rectangular coordinate as shown in fig. 2

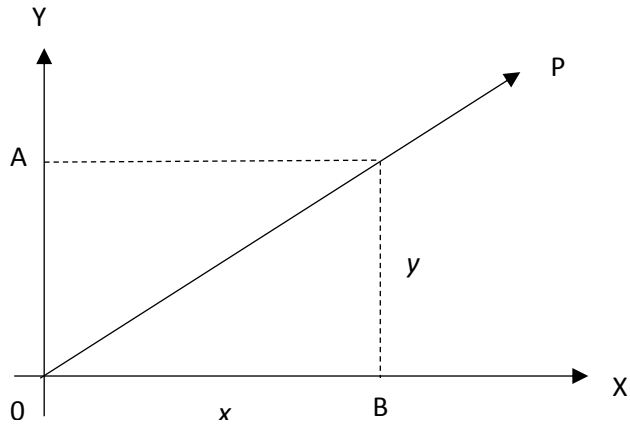


Fig. 2. Rectangular coordinates

From fig. 2, the position of point can be obtained by drawing PA and PB through the reference line OX and OY respectively. Then, measure OA and OB hence, the position of P is determined by the numerical values of OA and OB.

- ii. Choose any 2 perpendicular lines which is called the reference lines and their point of intersection is called the origin of rectangular coordinates. The description of the motion of the particle P can be done in terms of rectangular polar coordinate as shown in fig. 3

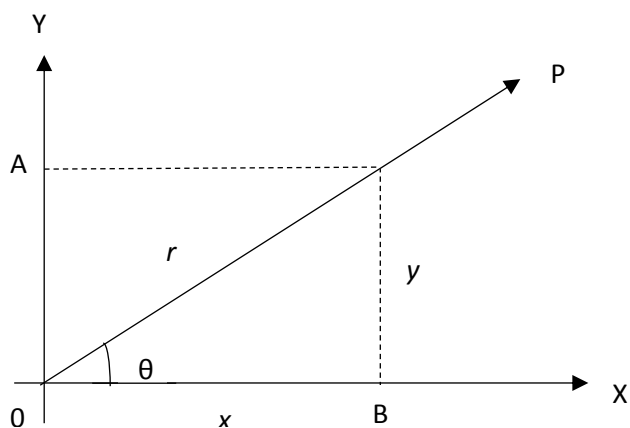


Fig.3. Rectangular polar coordinates

From fig. 3, the term of polar coordinates angle θ is the angle between the radius vector r , and the x-axis. It is measured in counter clockwise direction.

The radius vector r , is described by these equations below.

- $r^2 = x^2 + y^2$
- $\tan \theta = \frac{y}{x}$
- $x = r \cos \theta$
- $y = r \sin \theta$

UNITS AND DIMENSION

All physical quantities in science and engineering consists of 2 parts namely are:

1. Numerical value or numbers
2. Units which specifies the nature of the physical quantities.

Units: are standardized symbols of measurement of a physical quantity. There are 2 types of units namely

- Fundamental units
 - Derived units
- a. **Fundamental units:** are the basic units for physical quantity from which every other units can be generated. They are known as the S.I units which are formed from the M.K.S system of measurement. The M.K.S system are Metre (m), Kilogram (kg) and Seconds (s). These mentioned 3 basic units are the fundamental units.
 - b. **Derived units:** are units obtained from the symbols combination of the fundamental units e.g. speed (ms^{-1}), velocity (ms^{-1}), acceleration (ms^{-2}), density (kgm^{-3}) etc.

Dimension: is the numerical values that carry units. They form the backbone of all scientific observations. Similarly, any quantity in science and engineering that have units is known as dimensional quantity e.g. length, breadth, area, volume etc.

Dimensions are usually designated by the means of square brackets e.g. Distance = [Length] = [L].

Examples.

Since, all the units are in the S.I unit of measurement are fixed to the primary standard of length, time, and mass. The dimensions are:

Length = [L]

Mass = [M]

Time = [T]

i.e. The dimension of Area = length x breadth = [L] x [L] = [L²] or [L]² and the unit of Area = m²

DIMENSIONAL ANALYSIS

It express a particular physical quantity of its fundamental dimension. The dimension of any physical quantities are obtained from the definitions or units of the quantity. They are the integral part of the quantity. Note that, all dimension analysis have the following set of rules.

- I. The dimension determines the types of unit in a physical quantity. E.g. Suppose a given quantity has dimension $\frac{[L]}{[T]}$ then, the unit of the quantity must always be $\frac{[L]}{[T]} =$

m/s or km/hr .

- II. Dimension and units have algebraic examples;

$$\frac{kgm^2}{s} \times \frac{1}{kg^2 s/m^2} = \frac{kgm^2}{s} \times \frac{m^2}{kg^2 s} = \frac{m^4}{kg s^2}$$

- III. All equations must be homogeneous in the dimension i.e. “an equally reach” implies a comparison of both sides of the equation. Hence, both sides of an equation must have the same dimension. This rule is very important in checking equations of algebraic e.g. $A = B$.

$$[L][T] = [L][T]$$

Worked examples

1. Find the dimension of the equation: $x = x_0 + V_{x_0}t + \frac{1}{2}a_{x_0}t^2$ where x_0 = length,

V = velocity, a = acceleration and t = time

Soln.

Applying rule III, R.H.S = L.H.S

L.H.S; $x = [L]$

R.H.S; $x_0 = [L]$

$$V_{x_0}t = \frac{[L]}{[T]} \times [T] = [L]$$

$$\frac{1}{2}a_{x_0}t^2 = \frac{1}{2} \frac{[L]}{[T^2]} \times [T^2] = \frac{1}{2}[L]$$

$$x_0 + V_{x_0}t + \frac{1}{2}a_{x_0}t^2 = [L] + [L] + \frac{1}{2}[L] = \frac{5}{2}[L]$$

$$x = x_0 + V_{x_0}t + \frac{1}{2}a_{x_0}t^2 \rightarrow [L] = \frac{5}{2}[L]$$

$[L] = \frac{5}{2}[L]$ Since, $\frac{5}{2}$ is a numerical value then $[L] = [L]$ is correct dimensionally.

- IV. Units may be converted to other units of the same dimension. E.g. unit of seconds to minutes, metre to kilometre, grams to kilograms etc.

Worked examples:

1. Show that $E = mc^2$ is correctly dimensional where $E = \text{workdone}$; and $c = \text{speed of light}$.

Soln.

$$E = mc^2$$

$$\text{L.H.S; } E = \text{workdone} = \text{Force} \times \text{distance} = ma \times x$$

$$E = \frac{[M] \times [L] \times [L]}{[T^2]} = \frac{[M][L^2]}{[T^2]}$$

$$\text{R.H.S; } mc^2$$

$$m = \text{mass} = [M]$$

$$c^2 = \text{speed of light} = \frac{[L]^2}{[T]^2}$$

$$mc^2 = \frac{[M][L]^2}{[T]^2}$$

According to rule 3: $L.H.S = R.H.S$

$$E = \frac{[M][L^2]}{[T^2]} \text{ are the same (correctly dimensional).}$$

2. The period of vibration, t of a liquid drop is given by $t = kr^a\gamma^b\rho^c$, where k is a constant, r is the radius of the drop, ρ is the density of the liquid and γ is the surface tension. By dimension, (i) Find the values of the indices a , b and c .
(ii) Write the relation for t .

Soln.

$$t = kr^a \gamma^b \rho^c$$

Using dimensional analysis rule 3;

$$\text{L.H.S: } t = \text{period or time} = [T]$$

$$\text{R.H.S: } kr^a \gamma^b \rho^c = k \times [L]^a \times \left(\frac{f}{l}\right)^b \times \left(\frac{m}{l^3}\right)^c$$

$$\text{But } f = ma = [M][LT^{-2}]$$

$$kr^a \gamma^b \rho^c = k[L]^a \times \left(\frac{[M][LT^{-2}]}{[L]}\right)^b \times \left[\frac{M}{L^3}\right]^c$$

$$= k[L]^a \times ([M][T^{-2}])^b [ML^{-3}]^c$$

$$= k[L]^a [M^b][T^{-2}]^b [M^c][L^{-3}]^c$$

$$\text{R.H.S; } kr^a \gamma^b \rho^c = k[L]^{a-3c} [M]^{b+c} [T]^{-2b}$$

Compare the L.H.S to the R.H.S. i.e. $L.H.S = R.H.S$

$$[T] = k[L]^{a-3c} [M]^{b+c} [T]^{-2b}$$

$$1 \times 1 \times [T] = k[L]^{a-3c} [M]^{b+c} [T]^{-2b}$$

$$[M]^0 \times [L]^0 \times [T] = k[M]^{b+c} [L]^{a-3c} [T]^{-2b}$$

$$[M]^0 [L]^0 [T]^1 = k[M]^{b+c} [L]^{a-3c} [T]^{-2b}$$

The simultaneous equations are as follows;

$$[T]^1 = [T]^{-2b}$$

$$\text{Therefore, } 1 = -2b \quad (1)$$

$$[L]^0 = [L]^{a-3c}$$

$$\text{Therefore, } 0 = a - 3c \quad (2)$$

$$[M]^0 = [M]^{b+c}$$

$$\text{Therefore, } 0 = b + c \quad (3)$$

$$\text{From eqn. (1), } 1 = -2b \Rightarrow b = -1/2 \quad (4)$$

$$\text{From eqn. (2), } 0 = a - 3c \Rightarrow a = 3c \quad (5)$$

$$\text{From eqn. (3), } 0 = b + c \Rightarrow b = -c \quad (6)$$

Put (4) into (6) we have;

$$b = -c$$

$$c = -b = -(-1/2) = 1/2$$

$$c = 1/2$$

Put the value of c into (5), we have;

$$a = 3c = 3(1/2) = 3/2$$

$$a = 3/2$$

Therefore, $a = 3/2$, $b = -1/2$ and $c = 1/2$ substitute into the main dimensional equation to find $k = ?$

$$[M]^0[L]^0[T]^1 = k[M]^{b+c}[L]^{a-3c}[T]^{-2b}$$

$$[M]^0[L]^0[T]^1 = k[M]^{-1/2+1/2}[L]^{3/2-3(1/2)}[T]^{-2(-1/2)}$$

$$[T]^1 = k[M]^0[L]^0[T]^1$$

$$[T]^1 = k[T]^1 \quad (*)$$

$$k = \frac{[T]^1}{[T]^1} = 1$$

From the eqn. (*), $[T]^1 = k[T]^1$ where $k = 1$

$$[T]^1 = 1[T]^1$$

$$[T]^1 = [T]^1$$

$[T] = [T]$ Dimensionally correct so therefore, $t = kr^a\gamma^b\rho^c$ becomes

$$t = 1r^{3/2}\gamma^{-1/2}\rho^{1/2} = r^{3/2}\gamma^{-1/2}\rho^{1/2}$$

$$t = r^{3/2}\gamma^{-1/2}\rho^{1/2}$$

Exercise (UNITS & DIMENSION)

1. The displacement of a particle moving under uniform acceleration is some function of the elapsed time and the acceleration. Suppose we write this *displacement*

$$s = ka^m t^n$$

where k is a *dimensionless constant*. Show by dimensional analysis that this expression is satisfied if $m = 1$ and $n = 2$. Can this analysis give the value of k ?

2. The period T of a simple pendulum is measured in time units and is described by

$$T = 2\pi \sqrt{\frac{l}{g}}$$

where l is the *length* of the pendulum and g is the *freefall acceleration* in units of *length* divided by the *square of time*. Show that this equation is dimensionally correct.

3. Which of the equations below are dimensionally correct?

a) $v = v_0 + ax$

b) $y = (2m) \cos(kx)$, where $k = 2m^{-1}$

c) $v = s + at$

d) $V^2 = U^2 \sin^2 \theta - 2gs$

4. Newton's law of universal gravitation is represented by: $F = \frac{GMm}{r^2}$

Here F is the *gravitational force*, M and m are *masses*, and r is a *length*. Force has the S.I unit kgm/s^2 . What are the SI units of the *proportionality constant* G ?

5. Show that $E = P.E + K.E = mgh + \frac{1}{2}mV^2$ is correctly dimensional where $E =$ *workdone* in *Joules*.

6. Show that $T = \frac{2U \sin \theta}{g}$ is correctly dimensional where $T =$ *time of flight* in *seconds*, $U =$ *initial velocity* in *m/s*, $\theta =$ *elevation* and $g =$ *acceleration due to gravity* in *m/s*².

7. Show that $V^2 = U^2 + 2as$ is correctly dimensional where $U =$ *initial velocity* in *m/s*, $V =$ *final velocity* in *m/s*, $a =$ *acceleration* in *m/s*² and $s =$ *displacement* in *m*.

8. Derive the unit and dimension of Young modulus.

9. A particle of mass, m attached to a string of length, r ins whirled round in a circle with a speed, V . Use dimension analysis to find the form of the expression for the tension in the string.

10. If the distance (s) moved by a body is given by the equation; $S = Pt + Q \sin \theta + Rt^2 \cos \theta$, find the unit of P , Q and R appearing in the expression.

11. The radius, r of a circle inscribed in any triangle whose sides are a , b and c is given by;

$$r = \left[\frac{(s-a)(s-b)(s-c)}{s} \right]^{1/2}$$

Where s is an abbreviation for $\frac{(a+b+c)}{2}$. Check this formula for dimensional consistency.

12. What is the dimension of the physical quantity β in the equation below;

$$\text{Pressure} = \frac{\text{density}}{\beta}$$

13. Determine if the formula $T = 2\pi \sqrt{\frac{m}{x/F}}$ is dimensionally correct, where T is the

time/period, m is *mass*, F is *force* and x is *distance*.

14. By method of dimension show that the following equations are homogeneous

i. $F = k \frac{q_1 q_2}{r^2}$

ii. $H = \frac{v_0^2 \sin^2 \beta}{2g}$

15. The volume V of liquid that flows through a pipe in time t is given as;

$$\frac{V}{t} = \frac{\pi P r^4}{8Cl}$$

Where P is the pressure difference between the ends of the pipe and r is the radius. The constant C depends on the frictional effect of the liquid. Determine the base units of C .

16. A body moving through air at a high speed V experience a retarding force F given by;

$F = KA\rho V^x$, where A is the surface area of the body, ρ is the density of air and K is a numerical constant. Deduce the value of x .

17. Evaluate α and β in the equation below

$E = C m^\alpha v^\beta$, where E is the kinetic energy, m is mass, v is velocity and C is a dimensionless constant.

18. When a body is moving through a resisting medium such as air it experiences a drag force D which opposes the motion. D is given by the expression;

$$D = \frac{1}{2} C \rho A V^2$$

Where ρ is the density of the resisting medium, A is the effective cross-sectional area of the body, i.e. that area perpendicular to its velocity. C is called the drag coefficient. Show that C has no dimensions.

19. The equation relating current I through a semiconductor diode to the applied potential difference V at temperature T is;

$$I = I_0 e^{-eV/kT}$$

Where e in the eV is the electron charge and k is the Boltzmann constant. What are the dimensions of K ?

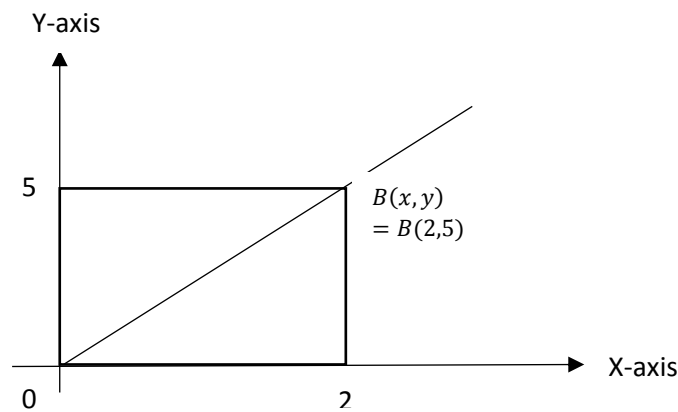
20. The velocity of waves on an ocean of density ρ is written as $V = A g^x \rho^y \lambda^z$, where A , x , y , z are constant, g is the acceleration due to gravity and λ is the wavelength of the wave. Using dimensional analysis, find the exact values of x , y and z .

2. RECTILINEAR MOTION: VELOCITY & ACCELERATION

Motion in Physics is the change in the position or orientation of a body with time. Motion along a line or a curve is called Translational motion. Motion that changes the orientation of a body is called Rotational motion. All motions are relative to some frame of reference.

Terminologies:

- Position: is referred to as the point in which an object can be located. The position of an object on a plane can be given by its coordinates. The x-coordinate is called *Abcissa* while the y-coordinate is called *Ordinate*. Thus, the location of an object at a point in a space (x-y plane) is regarded as the position of that object.



- Distance, s : is the gap between any two (2) positions in space. It is denoted by S and measured in metre (m). it is a scalar quantity.
- Displacement, s : is the distance covered in a specified direction. It is a vector quantity, also measured in metre (m).
- Speed: is defined as the time rate of change in distance moved in an unspecified direction. It is measured in m/s and also a scalar quantity.

$$Speed = \frac{distance}{time} (m/s)$$

- Velocity: is defined as the time rate of change in displacement. It is measured in m/s and also a vector quantity.

$$Velocity = \frac{displacement}{time} (m/s)$$

1. Uniform velocity: is the type of velocity made when equal distances are covered in equal time interval in the same direction.
2. Variable velocity: this is the velocity with which unequal distances are covered in equal interval of time in the same direction.
3. Average velocity: is also called mean velocity of a body. It is the velocity with which the distance travelled by the body in the same time interval is the same as that with the variable velocity.

$$Av. velocity = \frac{v + u}{2}$$

4. Instantaneous velocity: is the velocity of a body at a particular instant of time.

$$\text{i.e. } V = \frac{dx}{dt}$$

Note that when a body is thrown vertically upwards is instantaneously at rest at maximum height and the final velocity v as it attained its maximum height is zero.

5. Relative velocity: is the velocity of an object relative to a specific frame of reference.
- Acceleration: is the time rate of change in velocity. It is measured in m/s^2 .

$$Acceleration, a = \frac{\Delta \text{ in velocity}}{\text{time}} = \frac{dv}{dt}$$

When the velocity is increasing, the acceleration is reckoned as positive, when decreasing as negative (deceleration).

1. Uniform Acceleration: is the acceleration of a body which increases its velocity by constant amounts in equal intervals of time, no matter how small the time may be.
2. Instantaneous Acceleration: is the acceleration of the body at a particular point in

$$\text{time. i.e. } a = \frac{dv}{dt}$$

Worked examples:

1. The position of a particle moving along x-axis is given by $x = 2t^3 - 6t + 5$, where x is in metre and t in seconds. Find the distance travelled by the particle at (a) $t = 2s$ (b) $t = 5s$.

Soln.

$$x = 2t^3 - 6t + 5$$

(a) At time $t = 2s$,

$$x = 2(2)^3 - 6(2) + 5 = 9 \text{ m}$$

(b) At time $t = 5s$,

$$x = 2(5)^3 - 6(5) + 5 = 250 - 30 + 5 = 225 \text{ m}$$

2. The position of an object moving along x-axis is given by $x(t) = 3t^3 - 4t^2 + 6t$, where x is in metre and t is in seconds. What is the object's displacement and average velocity at the time interval from $t = 3s$ to $t = 5s$?

Soln.

$$x(t) = 3t^3 - 4t^2 + 6t$$

$$x(3) = 3(3)^3 - 4(3)^2 + 6(3) = 81 - 36 + 18 = 63 \text{ m}$$

$$x(5) = 3(5)^3 - 4(5)^2 + 6(5) = 375 - 100 + 30 = 305 \text{ m}$$

$$\text{Displacement } x(t) = x(5) - x(3) = 305 - 63 = 242 \text{ m}$$

$$\text{Average velocity } v(t) = \frac{dx}{dt} = \frac{x(5)-x(3)}{5-3} = \frac{305-63}{2}$$

$$v(t) = 121 \text{ ms}^{-1}$$

3. A particle moves along the x-axis from the origin, the magnitude of its position in metres is

$$x(t) = 10t^3 + 12t^2 + 5t + 8. \text{ Find the instantaneous acceleration at time } 3s.$$

Soln.

$$x(t) = 10t^3 + 12t^2 + 5t + 8$$

$$v(t) = \frac{dx}{dt} = 30t^2 + 24t + 5$$

$$a(t) = \frac{dv}{dt} = 60t + 24$$

At $t = 3s$,

$$a(t) = \frac{dv}{dt} = 60(3) + 24 = 204 \text{ ms}^{-2}$$

4. The acceleration of a particle is given by $a = 2t^2$. If the particle is at rest at $t = 0$, find the velocity and acceleration at $t = 2.5s$

Soln.

Given that $a = 2t^2$

$$v = \int a \, dt = \int 2t^2 \, dt = \frac{2t^3}{3} + c \dots\dots\dots (i)$$

$$v = 0 \text{ at } t = 0$$

$$\text{Substitute into eqn. (i) } 0 = \frac{2(0)^3}{3} + c \text{ or } c = 0$$

$$v = \frac{2t^3}{3}$$

At $t = 2.5s$,

$$v = \frac{2 \times (2.5)^3}{3} = \frac{31.25}{3} = 10.42 \text{ ms}^{-1}$$

$$\text{Recall that } a = 2t^2 = 2(2.5)^2 = 12.5 \text{ ms}^{-2}$$

VELOCITY – TIME GRAPH

A graph is the pictorial relationship between two varying quantities. The velocity-time graph is a plot of velocity against time which yields graph of various forms and it is more useful in giving accurate information concerning the motion of objects. The following information can be obtained from the graph;

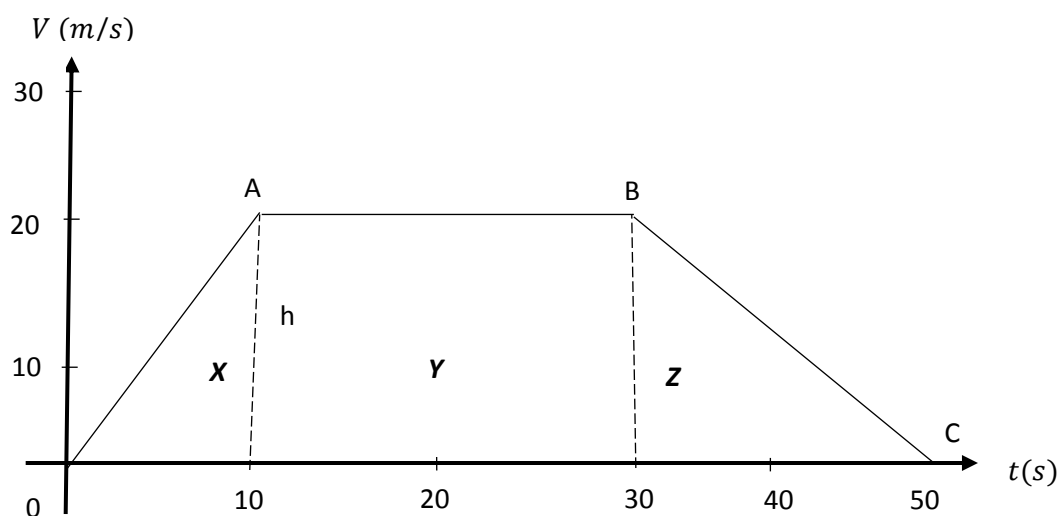
- (i) Acceleration (ii) Retardation/deceleration (iii) Distance covered (iv) Average speed.

Worked examples:

1. A motor car accelerates for 10 seconds to attain a velocity of 20 ms^{-1} . It continues with a uniform velocity for a further 20 seconds and then, decelerates so that it stops in 20 seconds.

Calculate (i) Acceleration (ii) Retardation/deceleration (iii) Distance covered.

Soln.



i. Acceleration, $a = \frac{v-u}{t} = \frac{20-0}{10} = \frac{20}{10} = 2 \text{ ms}^{-2}$

ii. Deceleration $\frac{v-u}{t} = \frac{0-20}{20} = \frac{-20}{20} = -1 \text{ ms}^{-2}$

iii. Distance travelled = Total area = Area of a triangle +

$$\text{Area of rectangle} + \text{Area of triangle} = \frac{1}{2}(b \times h) + (l \times b) + \frac{1}{2}(b \times h)$$

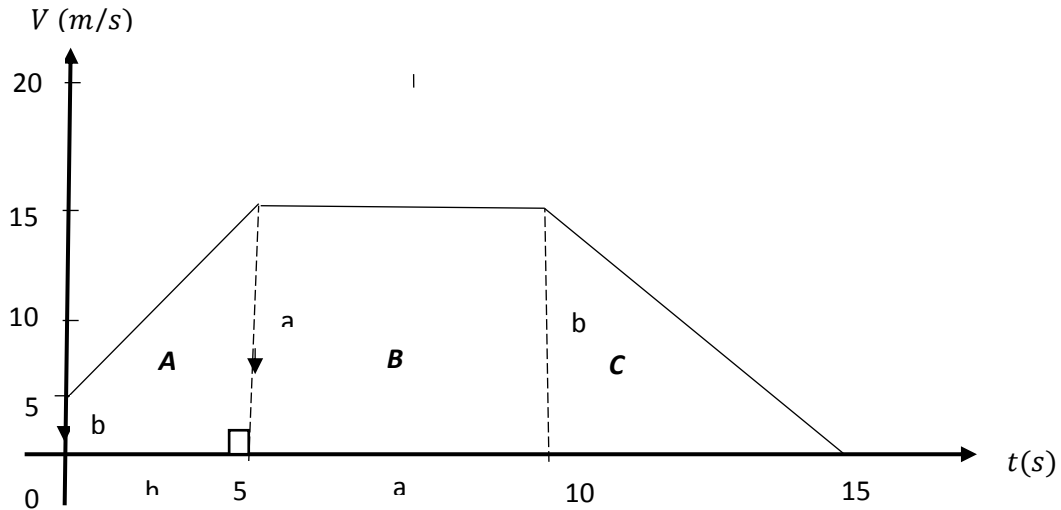
$$= \frac{1}{2}(10 \times 20) + (20 \times 20) + \frac{1}{2}(20 \times 20) = 100 + 400 + 200 = 700 \text{ m}$$

$$\therefore \text{The distance travelled} = 700 \text{ m}$$

OR

$$\text{The distance travelled} = \frac{1}{2}h(AB + OC) = \frac{1}{2} \times 20(20 + 50) = 700 \text{ m}$$

2.



Calculate (i) Acceleration (ii) Total distance covered (iii) Retardation/deceleration (iv) Average speed.

Soln.

i. Acceleration, $a = \frac{v-u}{t} = \frac{15-5}{5} = \frac{10}{5} = 2 \text{ ms}^{-2}$

ii. Total distance covered = *Total area*

= *Area of a trapezium, A + Area of rectangle, B + Area of triangle, C*

$$= \frac{1}{2}(a+b)h + (a \times b) + \frac{1}{2}(b \times h)$$

$$= \frac{1}{2}(5+15)5 + (5 \times 15) + \frac{1}{2}(5 \times 15) = 50 + 75 + 37.5 = 162.5 \text{ m}$$

$\therefore$ The total distance covered = 162.5 m

iii. Deceleration = $\frac{v-u}{t} = \frac{0-15}{5} = \frac{-15}{5} = -3 \text{ ms}^{-2}$

iv. Average speed = $\frac{\text{total distance}}{\text{time-taken}} = \frac{162.5}{15} = 10.83 \text{ ms}^{-1}$

Analysis of Rectilinear motion (Equation of Motion):

Consider a body with an initial velocity u , moving in time t , let its final velocity be v and its acceleration be a ;

$$\begin{aligned} \text{Acceleration, } a &= \frac{v - u}{t} \\ at &= v - u \\ v &= u + at \end{aligned} \tag{1}$$

$$\text{Distance travelled, } s = \text{Average velocity} \times \text{time} = \frac{(u + v)}{2} \times t$$

$$s = \frac{(u+v)}{2} \times t \tag{2}$$

Since $v = u + at$, substitute into eqn. (2)

$$\begin{aligned} s &= \frac{(u + u + at)}{2} \times t \\ s &= \frac{(ut + ut + at^2)}{2} = \frac{2ut + at^2}{2} \\ s &= \frac{2ut}{2} + \frac{at^2}{2} = ut + \frac{1}{2}at^2 \\ s &= ut + \frac{1}{2}at^2 \end{aligned} \tag{3}$$

From eqn. (1), $v = u + at$, square both sides.

$$\begin{aligned} v^2 &= (u + at)^2 = u^2 + 2uat + (at)^2 \\ v^2 &= u^2 + 2uat + (at)^2 \\ v^2 - u^2 &= 2uat + a^2t^2 \\ v^2 - u^2 &= 2a(2ut + \frac{1}{2}at^2) \end{aligned}$$

But from eqn. (2) $s = ut + \frac{1}{2}at^2$, substitute in the above eqn.

$$\begin{aligned} v^2 - u^2 &= 2a(s) \\ v^2 - u^2 &= 2as \\ v^2 &= u^2 + 2as \end{aligned} \tag{4}$$

3. A man throws a ball vertically upward with an initial speed of 20 ms^{-1} . What is the maximum height reached by the ball and how long does it take to return to the point it was thrown?

Soln.

$$u = 20 \text{ ms}^{-1}, g = 9.81 \text{ ms}^{-2}, s = ?$$

At maximum height, $v = 0$

$$v^2 = u^2 + 2as$$

Where $a = -g$

$$v^2 = u^2 - 2gs = 20^2 - 2(9.81)s = 0$$

$$19.62s = 400$$

$$\therefore s = \frac{400}{19.62} = 20.4 \text{ m}$$

PROJECTILE MOTION:

Projectile motion is the motion of a body on a parabolic path. Projectile motion is a form of motion experienced by an object or particle (projectile) that is thrown near the earth's surface and moves along a curved path (trajectory) under the action of gravity only. (Assuming the effects of air resistance is negligible).

Examples: motion of a long jumper, motion of a missile warhead, motion of a javelin, motion of a stone from a catapult etc.

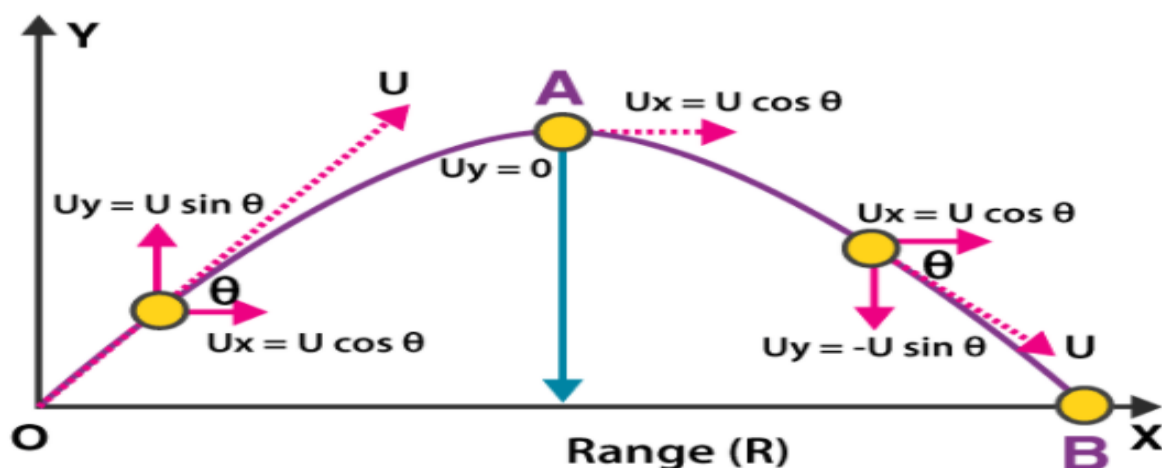


Fig. 1. Trajectory of a Projectile.

The velocity, U of a projectile can be resolved into two (2) components namely are:

- Horizontal component, $U_x = U \cos \theta$
- Vertical component, $U_y = U \sin \theta$

Also, at maximum height, $U = 0$ but the object is thrown vertically upward $V = 0$.

1. Time to taken (t) is the time taken to reach the maximum height. It can be obtained using the 1st equation of motion under gravity; $v = u + at$

$$v = U_y + at$$

Where $v = 0$, $a = -g$

$$0 = U_y - gt$$

$$gt = U_y$$

$$t = \frac{U_y}{g} = \frac{U \sin \theta}{g}$$

$\therefore$ the time, t used by the projectile body to reach a maximum height is $t = \frac{U \sin \theta}{g}$ (1)

2. Time of flight (T) is the total time used by the object in air. It is the time taken by the body to return to the plane of projection from the instant of inception of being projected.

$$T = 2t$$

$$T = 2 \times \frac{U \sin \theta}{g} = \frac{2U \sin \theta}{g}$$

$$\therefore T = \frac{2U \sin \theta}{g} \quad (2)$$

3. Maximum height attained (H_{max}) is the maximum vertical distance travelled by a projected body. It is denoted by H .

Use the eqn. $v^2 = u^2 + 2as$ where $v = 0$, $a = -g$ and $s = H$

$$v^2 = U_y^2 - 2gH$$

$$0 = U_y^2 - 2gH$$

$$U_y^2 = 2gH$$

$$H = \frac{U_y^2}{2g} = \frac{(U \sin \theta)^2}{2g} = \frac{U^2 \sin^2 \theta}{2g}$$

$$\therefore H = \frac{U^2 \sin^2 \theta}{2g} \quad (3)$$

4. Range (R) is the total horizontal distance travelled by the body from the point where the body hits the projection plane.

Use the eqn. $s = ut + \frac{1}{2}at^2$ where $a = g = 0$, $u = U_x$, $t = T$ and $s = R$

$$s = ut + \frac{1}{2}gt^2 = ut = U_x T = U \cos \theta \times T$$

$$R = U \cos \theta \times T = U \cos \theta \times \frac{2U \sin \theta}{g}$$

where, $T = \frac{2U \sin \theta}{g}$

$$R = U \cos \theta \times \frac{2U \sin \theta}{g} = \frac{2U^2 \sin \theta \cos \theta}{g}$$

$$R = \frac{2U^2 \sin \theta \cos \theta}{g}$$

Recall that $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\therefore R = \frac{U^2 \sin 2\theta}{g} \quad (4)$$

Maximum range (R_{max}) is obtained in the horizontal plane when the angle the projected object makes with the horizontal plane is 45° .

$$R_{max} = \frac{U^2 \sin 2(45^\circ)}{g} = \frac{U^2 \sin 90^\circ}{g}$$

$$R_{max} = \frac{U^2(1)}{g} = \frac{U^2}{g}$$

$$\therefore R_{max} = \frac{U^2}{g} \quad (5)$$

Worked examples

1. A stone is shot-out from a catapult with initial velocity of 30 m/s at an elevation of 60°. Find (i) the time of flight (ii) maximum height (iii) the range

Soln.

- i. Time of flight $T = \frac{2U \sin \theta}{g}$

$$T = \frac{2(30) \sin 60}{10} = 5.20 \text{ s}$$

- ii. Maximum height $H = \frac{U^2 \sin^2 \theta}{2g}$

$$H = \frac{(30)^2 \sin^2 60}{2(10)} = 33.8 \text{ m}$$

- iii. Range $R = \frac{U^2 \sin 2\theta}{g}$

$$R = \frac{(30)^2 \sin 2(60)}{10} = 78 \text{ m}$$

2. A soccer player kicks a ball at an angle of 37° to the horizontal plane with an initial speed of 50 m/s. Assume that the ball moves in a vertical plane and that air resistance is negligible. Find

- i. The time at which the ball reaches the highest point of its trajectory.
- ii. How high does the ball go?
- iii. What is the range of the ball?
- iv. How long is it in air?
- v. What is the velocity of the ball as it strikes the ground?

Soln.

- i. The time at which the ball reaches the highest point of its trajectory, $t = \frac{U \sin \theta}{g}$

$$t = \frac{50 \sin 37^\circ}{10} = 3.01 \text{ s}$$

ii. Height of the ball, $H = \frac{U^2 \sin^2 \theta}{2g} = \frac{50^2 \sin^2 37^\circ}{2(10)} = 45.27 \text{ m}$

iii. Range, $R = \frac{U^2 \sin 2\theta}{g} = \frac{50^2 \sin 2(37^\circ)}{10} = 240.32 \text{ m}$

iv. $T = 2t = 2 \times 3.01 = 6.02 \text{ s}$

v. The velocity of the ball as it strikes the ground, $v = u + at$. Note that, $U = 0$ when it strikes the ground and $t = T$.

$$v = aT = gT = 10 \times 6.02 = 60.20 \text{ m/s}$$

Exercise (RECTILINEAR MOTION)

1. A stone is thrown vertically upward with velocity of 15 m/s at the same time, 10 m vertically above a second stone is allowed to fall. After what time and at what height do they collide [take $g = 10 \text{ ms}^{-2}$]
2. A particle moves along the x -axis according to the equation $x = 2.00 + 3.00t - t^2$, where x is in meters, t is in seconds. At $t = 3.00 \text{ s}$, find (a) the position of the particle (b) its velocity and (c) its acceleration.
3. An object moves along the x -axis according to the equation;
 $x = (3.00t^2 - 2.00t + 3.00)\text{m}$. Determine;
 - (a) The average speed between $t = 2.00 \text{ s}$ and $t = 3.00 \text{ s}$.
 - (b) The instantaneous speed at $t = 2.00 \text{ s}$ and $t = 3.00 \text{ s}$.
 - (c) The average acceleration between $t = 2.00 \text{ s}$ and $t = 3.00 \text{ s}$ and
 - (d) The instantaneous acceleration at $t = 2.00 \text{ s}$ and $t = 3.00 \text{ s}$.
4. A particle motion is given by the equation $a = 2t^2 - 5t + 6$, where ' a ' is an acceleration in m/s^2 and ' t ' is time in seconds. The velocity and displacement after two seconds were 10.5 m/s and 17.8 m respectively.

- i. Find the velocity and displacement at the time of one second and three seconds.
 - ii. Draw the velocity – time graph for this motion.
5. The distance between two stations is 2.5 km , a locomotive starting from one station, gives the train an acceleration (reaching a speed of 20 km/hour in 2.5 minutes until the speed reaches 80 km/hr . this speed is maintained until brakes are applied and the train is brought to rest at the second station under a negative acceleration of 0.9 m/s^2 . Find the time taken to perform this journey.
 6. An object is projected with a velocity of 100 m/s at an angle 60° to the vertical. Calculate the time taken by the object to reach the highest point (take $g = 10\text{ m/s}^2$).
 7. A stone of mass 0.7 kg is projected vertically upwards with a speed of 5 m/s . Calculate the maximum height reached. (Take $g = 9.8\text{ m/s}^2$).
 8. The coordinate of a particle in *meters* is given by; $(t) = 16t - 3.0t^3$, where the time t is in seconds. Determine the time at which the particle is momentarily at rest.
 9. A body of mass m moves along x -axis such that at time t its position is given by $x(t) = at^{3/2} - bt + c$ where a , b , and c are constants. (a) Calculate the acceleration of the body (b) the force acting on the body.
 10. A projectile is fired from the edge of a 150 m high cliff with an initial velocity of 180 m/s at an angle of elevation 30° with the horizontal. Neglecting air resistance, find (i) the horizontal distance from the gun to the point where the projectile strikes the ground, and (ii) the greatest elevation above the ground reached by the projectile.

3. SCALARS AND VECTORS

Scalar quantity is a quantity that is fully described by only magnitude but no direction. E.g. length, distance, area, volume, mass, energy, speed etc.

Vector quantity is a quantity that is fully described by both magnitude and direction. E.g. displacement, velocity, force, momentum, acceleration etc.

Vector symbols

A vector is represented in a diagram by an arrow or a directed line segment. The length of the arrow is directly proportional to the magnitude of the vector, and the direction is the same as the direction of the vector. The tail of the arrow is the initial point, and the head of the arrow is the final point. If the vector is drawn to scale, the scale should be indicated on the diagram.



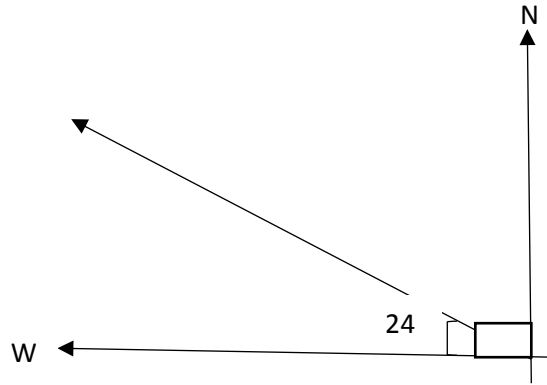
Fig. 1. Examples of vector quantities where (a) Displacement vector (scale: $1 \text{ cm} = 10 \text{ m}$) and (b) Acceleration vector (scale: $1 \text{ cm} = 5 \text{ m/s}^2$).

In this text, a vector quantity is indicated by an arrow above the letter representing the vector (e.g. $\vec{A}$, $\vec{a}$, $\vec{P}$, $\sum \vec{F}$ etc.). The magnitude of a vector is indicated by the absolute value symbol $|\vec{A}|$ or simply A . The magnitude is always positive (unless it is zero).

Directions of Vectors

The directions of vectors are indicated in square brackets following the magnitude and units of measurement. The four (4) compass directions: East, West, North and South are indicated as [E], [W], [N] and [S] respectively. Other examples are [down], [forward], [12.5° below the horizontal], [toward earth's centre] and [24° N of W].

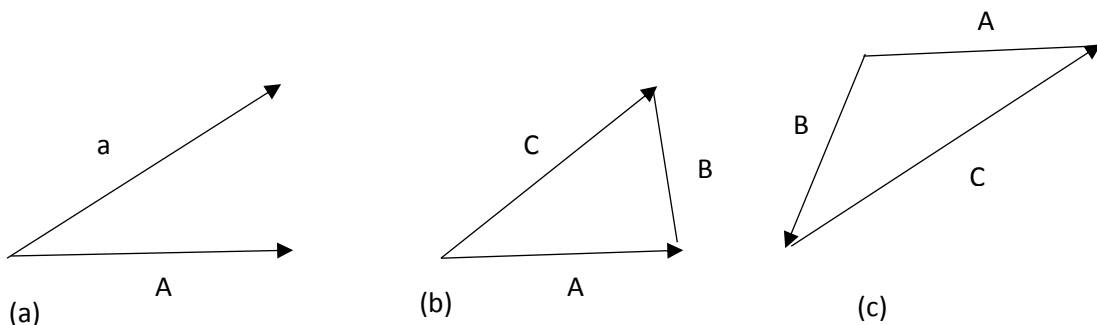
For instance, 24° N of W is simply 156° as shown in the figure below.



Resolution of Vectors

Is a method employed to simply add, subtract, multiply algebraically scalar and vector quantities and also to apply the parallelogram's or polygon's laws of vectors for all vector quantities.

1. Addition of vectors: Let a particle, P be influenced by two (2) vectors A and B . the resultant of these 2 vectors A and B , draws the vector A and draws the vector B in that order.



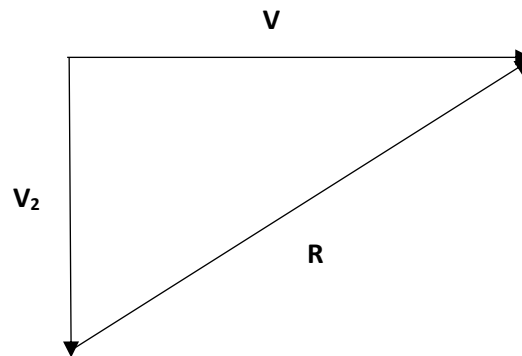
$$A + B = C$$

Also, the resultant of 2 vectors A & B ; $A + B = B + A$

The following are methods of vector addition;

- **The Pythagorean Theorem and trigonometric methods:** is a useful method for determining the result of adding two (and only two) vectors that make a right-angle to each other. The method is not applicable for adding more than two vectors or for adding vectors that are not at 90-degrees to each other. The

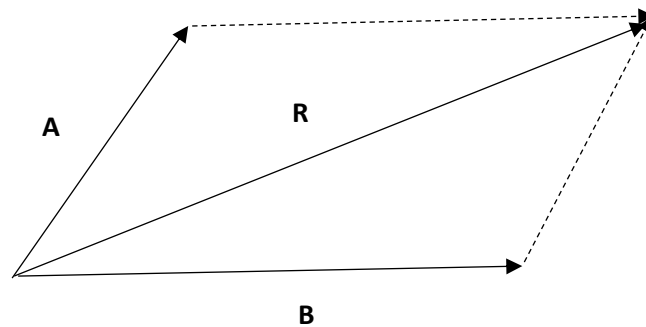
Pythagorean Theorem is a mathematical equation that relates the length of the sides of a right triangle to the length of the hypotenuse of a right triangle.



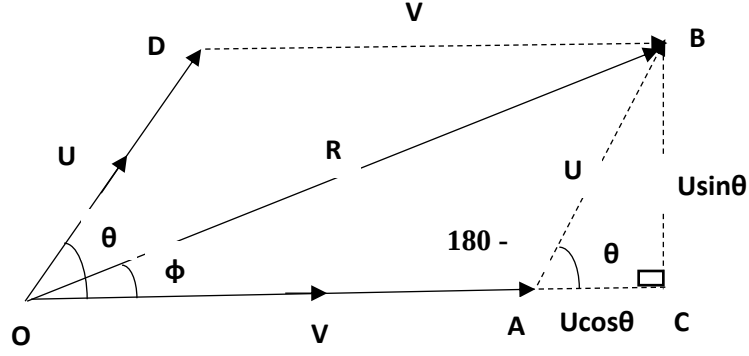
$$R^2 = V_1^2 + V_2^2$$

$$R = \sqrt{V_1^2 + V_2^2}$$

- **Parallelogram method:** the resultant of two vectors which inclined to each other may be represented by the diagonal of a parallelogram drawn with the two vectors as adjacent sides.



Parallelogram law of vectors states that if two vectors can be represented in magnitude and direction by the adjacent sides of a parallelogram, the diagonal of the parallelogram drawn from the point of intersection of the vectors represents the resultant vector in magnitude and direction.



Now expand **A** to **C** and draw **BC** perpendicular to **OC**.

$$\text{In } \triangle OCB, OB^2 = OC^2 + BC^2 = (OA + AC)^2 + BC^2 \quad (1)$$

$$\text{In } \triangle ABC, \cos \theta = \frac{AC}{AB} = \frac{AC}{U}$$

$$AC = U \cos \theta \quad (2)$$

$$\text{Also, } \sin \theta = \frac{BC}{AB} = \frac{BC}{U}$$

$$BC = U \sin \theta \quad (3)$$

Put (2) and (3) into (1) we have;

$$R^2 = (V + U \cos \theta)^2 + (U \sin \theta)^2 = (V^2 + U^2 \cos^2 \theta + 2UV \cos \theta) + U^2 \sin^2 \theta$$

$$R^2 = V^2 + U^2 \sin^2 \theta + U^2 \cos^2 \theta + 2UV \cos \theta = V^2 + U^2 (\sin^2 \theta + \cos^2 \theta) + 2UV \cos \theta$$

$$R^2 = V^2 + U^2(1) + 2UV \cos \theta$$

$$R^2 = V^2 + U^2 + 2UV \cos \theta$$

$$R = \sqrt{U^2 + V^2 + 2UV \cos \theta} \quad (4)$$

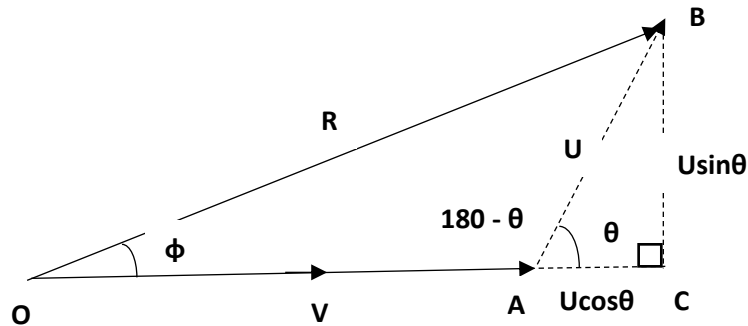
Note that $\cos \theta = -\cos(180 - \theta)$

$$R = \sqrt{U^2 + V^2 - 2UV \cos(180 - \theta)} \quad (5)$$

$$\text{In } \triangle BOC, \tan \phi = \frac{BC}{OC} = \frac{BC}{OA+AC} = \frac{U \sin \theta}{V+U \cos \theta}$$

$$\phi = \tan^{-1} \left[\frac{U \sin \theta}{V+U \cos \theta} \right] \quad (6)$$

Triangle law of vector (Force): if three forces acting at a point represent in magnitude and direction by the sides of a triangle taken in order they will be in equilibrium.

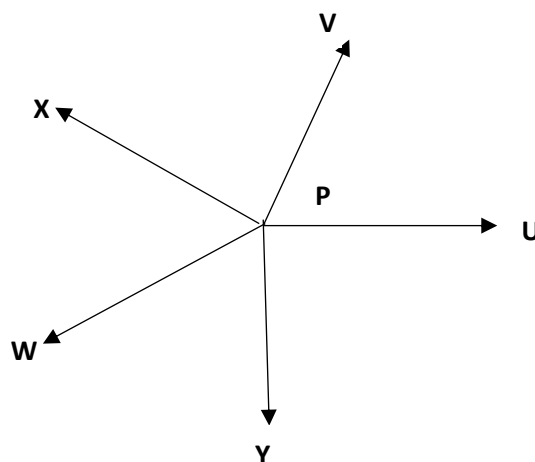


Where **R** = resultant velocity, **U** and **V** = 2 velocities.

$$R = \sqrt{U^2 + V^2 + 2UV \cos \theta} \text{ or } \sqrt{U^2 + V^2 - 2UV \cos(180 - \theta)}$$

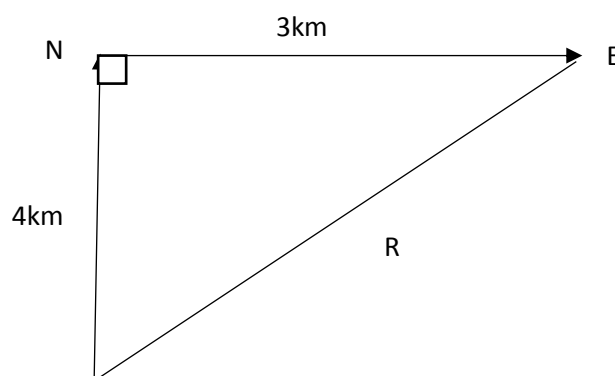
$$\phi = \tan^{-1} \left[\frac{U \sin \theta}{V + U \cos \theta} \right]$$

Polygon's law of velocities: states that if 2 or more particles simultaneously possess a number of uniform velocities in magnitude and direction by the sides of a polygon taken in order, the resultant velocity is represented in magnitude and direction by the closing side of the polygon taken in the opposite order.



Worked examples

1. A boy running in the northern direction later changed his direction and started running toward the eastern direction. If the distance covered in the northern direction is 4 km , and in the eastern direction is 3 km , determines (i.) the magnitude of his resultant vector (ii.) his direction.



Soln.

- i. $R^2 = P^2 + Q^2 = 4^2 + 3^2 = 25$

$$R^2 = 25$$

$$R = \sqrt{25} = 5\text{ km}$$

$\therefore$ The magnitude of his resultant is 5 km .

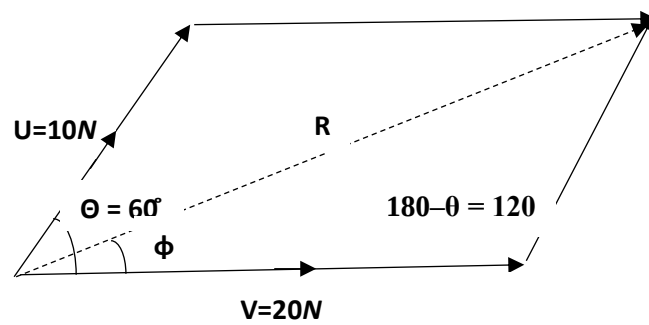
- ii. The direction; $\tan \theta = \frac{y}{x} = \frac{3}{4}$

$$\tan \theta = 0.75$$

$$\theta = \tan^{-1}(0.75) = 36.86^\circ$$

$$\therefore \theta = 36.86^\circ$$

2. Two forces 10 N and 20 N are inclined at angle 60° to each other. Find the resultant force and its angle α (direction).

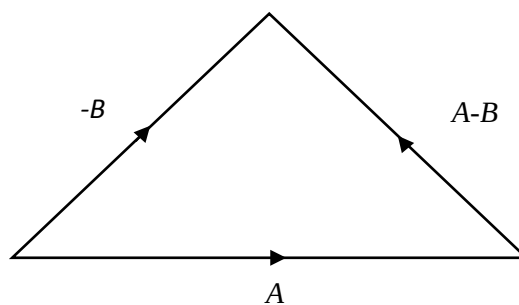


$$\begin{aligned} \text{i.} \quad R &= \sqrt{U^2 + V^2 + 2UV \cos \theta} \text{ or } \sqrt{U^2 + V^2 - 2UV \cos(180 - \theta)} \\ R &= \sqrt{10^2 + 20^2 + 2(10)(20) \cos 60^\circ} = \sqrt{100 + 400 + 400 \cos 60^\circ} \\ R &= \sqrt{700} = 26.46 \text{ N} \end{aligned}$$

$$\therefore R = 26.46 \text{ N}$$

$$\begin{aligned} \text{ii.} \quad \text{To determine the direction, } \alpha &= \tan^{-1} \left[\frac{U \sin \theta}{V + U \cos \theta} \right] \\ &= \tan^{-1} \left[\frac{10 \sin 60}{20 + 10 \cos 60} \right] = \tan^{-1} \left[\frac{8.66}{25} \right] = \tan^{-1}(0.3464) \\ \therefore \alpha &= 19.1^\circ \end{aligned}$$

2. Subtraction of vectors: the difference between 2 vectors A and B is obtained by taking the difference between the first vector and the second vector.



$$-B + A \neq A - B$$

3. Multiplication of vectors: multiplication of a vector by a scalar changes only the magnitude of the vector, but leaves its direction unchanged. The scalar changes the size of the vector. E.g.

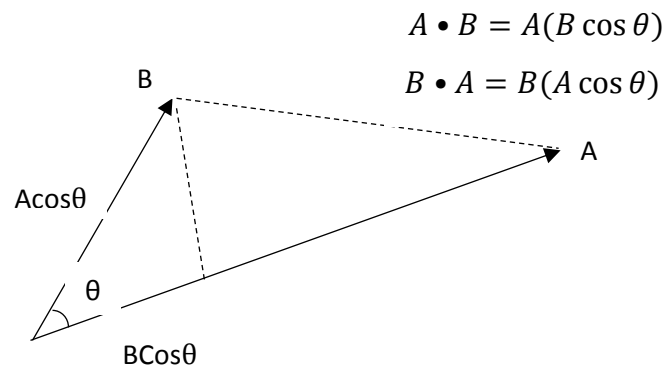
$$\vec{A} = 2i + 2j + 2k$$

Multiplied by the scalar 2 is;

$$2\vec{A} = 2(2i + 2j + 2k) = 4i + 4j + 4k$$

There are two (2) ways of multiplying a vector namely; Dot product and Cross product.

- **Dot Product:** the dot product of two (2) vectors is the magnitude of one times the projection of the second onto the first.



The symbol used to represent this operation is a small dot at middle height ($\bullet$), which is where the name “dot product” comes from. Since this product has magnitude only, it is also known as the Scalar product.

$$\vec{A} \bullet \vec{B} = |A||B| \cos \theta \quad (1)$$

The dot product is distributive; $A \bullet (B + C) = A \bullet B + A \bullet C$ and

Also, commutative; $A \bullet B = B \bullet A$

Since, the projection of a vector on to itself leaves its magnitude unchanged, the dot product of any vector with itself is the square of the vector’s magnitude.

$$\vec{A} \bullet \vec{A} = |A||A| \cos 0 = |A|^2(1) = |A|^2 = A^2 \quad (2)$$

Applying same rule for the dot product of any unit vector with itself is 1. In addition, since a vector has no projection perpendicular to itself, the dot product of any unit with any other vector is 0.

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1) \cos 0^\circ = 1 \quad (3)$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = (1)(1) \cos 90^\circ = 0 \quad (4)$$

Using this knowledge, we can derive a formula for the dot product of any two vectors in rectangular form.

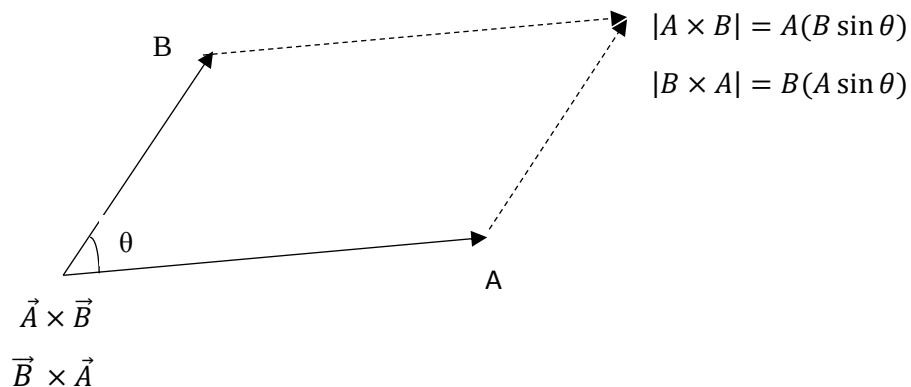
$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= A_x \hat{i} \cdot B_x \hat{i} + A_x \hat{i} \cdot B_y \hat{j} + A_x \hat{i} \cdot B_z \hat{k} + A_y \hat{j} \cdot B_x \hat{i} + A_y \hat{j} \cdot B_y \hat{j} + A_y \hat{j} \cdot B_z \hat{k} + A_z \hat{k} \cdot B_x \hat{i} \\ &\quad + A_z \hat{k} \cdot B_y \hat{j} + A_z \hat{k} \cdot B_z \hat{k} \end{aligned}$$

$$\vec{A} \cdot \vec{B} = A_x B_x (\hat{i} \cdot \hat{i}) + A_y B_y (\hat{j} \cdot \hat{j}) + A_z B_z (\hat{k} \cdot \hat{k})$$

$$\therefore \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z \quad (5)$$

$$\vec{A} \cdot \vec{A} = A_x A_x + A_y A_y + A_z A_z = A_x^2 + A_y^2 + A_z^2$$

- **Cross Product:** the cross – product of two vectors is the area of the parallelogram between them.



The symbol used to represent this operation is a large diagonal cross ($\times$), which is where the name “Cross product” comes from. Since, this product has magnitude and direction, it is also known as the Vector product.

$$\vec{A} \times \vec{B} = |A||B| \sin \theta \cdot \hat{n} \quad (6)$$

Where $|A|$ and $|B|$ are the magnitude of A and B .

$\hat{n}$ is the unit vector perpendicular to the plane formed by vector.

The cross-product is distributive i.e. $A \times (B + C) = (A \times B) + (A \times C)$ but not commutative i.e. $A \times B = -B \times A$.

Since, two identical vectors produce a degenerate parallelogram with no area, the cross product of any vector with itself is zero (0). i.e. $A \times A = 0$

Also, apply the above rule means that the cross-product of any unit vector with itself is zero.

$$\text{i.e. } \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = (1)(1) \sin 0^\circ = 0 \quad (7)$$

$$\hat{i} \times \hat{j} = \hat{j} \times \hat{k} = \hat{k} \times \hat{i} = (1)(1) \sin 90^\circ = 1 \quad (8)$$

Using this knowledge, we can also derive a formula for the cross-product of any two vectors in rectangular form.

$$\begin{aligned} \vec{A} \times \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= A_x \hat{i} \times B_x \hat{i} + A_x \hat{i} \times B_y \hat{j} + A_x \hat{i} \times B_z \hat{k} + A_y \hat{j} \times B_x \hat{i} + A_y \hat{j} \times B_y \hat{j} + A_y \hat{j} \times B_z \hat{k} + A_z \hat{k} \times B_x \hat{i} \\ &\quad + A_z \hat{k} \times B_y \hat{j} + A_z \hat{k} \times B_z \hat{k} \end{aligned}$$

$$\begin{aligned} \vec{A} \times \vec{B} &= A_x B_y (\hat{i} \times \hat{j}) + A_x B_z (\hat{i} \times \hat{k}) + A_y B_x (\hat{j} \times \hat{i}) + A_y B_z (\hat{j} \times \hat{k}) + A_z B_x (\hat{k} \times \hat{i}) \\ &\quad + A_z B_y (\hat{k} \times \hat{j}) \end{aligned}$$

$$\vec{A} \times \vec{B} = A_x B_y (\hat{k}) + A_x B_z (-\hat{j}) + A_y B_x (-\hat{k}) + A_y B_z (\hat{i}) + A_z B_x (\hat{j}) + A_z B_y (-\hat{i})$$

$$\vec{A} \times \vec{B} = A_x B_y (\hat{k}) - A_x B_z (\hat{j}) - A_y B_x (\hat{k}) + A_y B_z (\hat{i}) + A_z B_x (\hat{j}) - A_z B_y (\hat{i})$$

$$\therefore \vec{A} \times \vec{B} = (A_y B_z - A_z B_y) (\hat{i}) + (A_z B_x - A_x B_z) (\hat{j}) + (A_x B_y - A_y B_x) (\hat{k}) \quad (9)$$

There is a simpler way to write this. For those of you familiar with matrices, the cross-product of two vectors is the determinant of the matrix whose first row is the unit vectors, second row is the first vector and third row is the second vector.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = +\hat{i} \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - \hat{j} \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + \hat{k} \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$\therefore \vec{A} \times \vec{B} = (A_y B_z - A_z B_y)(\hat{i}) + (A_z B_x - A_x B_z)(\hat{j}) + (A_x B_y - A_y B_x)(\hat{k})$$

Worked examples

1. If $\vec{U} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{V} = \hat{i} + 2\hat{j}$. Find the following;

i. $\vec{U} \cdot \vec{V} = \vec{V} \cdot \vec{U}$

ii. The angle between the 2 vectors, θ .

iii. $\vec{U} \times \vec{V} = -\vec{V} \times \vec{U}$

Soln.

i. $\vec{U} \cdot \vec{V} = (2\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} + 2\hat{j} + 0\hat{k}) = 2\hat{i} \cdot \hat{i} + \hat{j} \cdot 2\hat{j} - \hat{k} \cdot 0\hat{k} = 2\hat{i}^2 + 2\hat{j}^2 - 0\hat{k}^2$

$$\vec{U} \cdot \vec{V} = 2(1) + 2(1) + 0 = 2 + 2 = 4$$

$$\vec{U} \cdot \vec{V} = 4$$

$$\begin{aligned} \vec{V} \cdot \vec{U} &= (\hat{i} + 2\hat{j} + 0\hat{k}) \cdot (2\hat{i} + \hat{j} - \hat{k}) = \hat{i} \cdot 2\hat{i} + 2\hat{j} \cdot \hat{j} + 0\hat{k} \cdot -\hat{k} \\ &= 2\hat{i}^2 + 2\hat{j}^2 - 0\hat{k}^2 \end{aligned}$$

$$\vec{V} \cdot \vec{U} = 2(1) + 2(1) + 0 = 2 + 2 = 4$$

$$\vec{V} \cdot \vec{U} = 4$$

$$\therefore \text{Therefore, } \vec{U} \cdot \vec{V} = \vec{V} \cdot \vec{U} = 4$$

ii. $\vec{U} \cdot \vec{V} = |\vec{U}||\vec{V}| \cos \theta$

$$\cos \theta = \frac{\vec{U} \cdot \vec{V}}{|\vec{U}||\vec{V}|}$$

$$\theta = \cos^{-1} \left[\frac{\vec{U} \cdot \vec{V}}{|\vec{U}| |\vec{V}|} \right]$$

$$|\vec{U}| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$|\vec{V}| = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{1 + 4 + 0} = \sqrt{5}$$

$$\vec{U} \cdot \vec{V} = 4$$

$$\theta = \cos^{-1} \left[\frac{4}{\sqrt{6}\sqrt{5}} \right] = \cos^{-1} \left[\frac{4}{\sqrt{30}} \right] = 43.09^\circ$$

$$\therefore \theta = 43.09^\circ$$

$$\text{iii. } \vec{U} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 0 \end{vmatrix} = +\hat{i} \begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$$

$$\vec{U} \times \vec{V} = \hat{i}[(1 \times 0) - (2 \times -1)] - \hat{j}[(2 \times 0) - (-1 \times 1)] + \hat{k}[(2 \times 2) - (1 \times 1)]$$

$$\vec{U} \times \vec{V} = \hat{i}[0 - (-2)] - \hat{j}[0 - (-1)] + \hat{k}[4 - 1] = \hat{i}[2] - \hat{j}[1] + \hat{k}[3]$$

$$= 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\therefore \vec{U} \times \vec{V} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$-\vec{V} \times \vec{U} = - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 2 & 1 & -1 \end{vmatrix} = -\hat{i} \begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix} + \hat{j} \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} - \hat{k} \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$$-\vec{V} \times \vec{U} = -\hat{i}[(2 \times -1) - (0 \times 1)] + \hat{j}[(1 \times -1) - (0 \times 2)] - \hat{k}[(1 \times 1) - (2 \times 2)]$$

$$= -\hat{i}[-2 - 0] + \hat{j}[-1 - 0] - \hat{k}[1 - 4] = -\hat{i}[-2] + \hat{j}[-1] - \hat{k}[-3] = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$-\vec{V} \times \vec{U} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\therefore \vec{U} \times \vec{V} = -\vec{V} \times \vec{U} = 2\hat{i} - \hat{j} + 3\hat{k}$$

2. Calculate a unit vector in the direction of $\vec{b} = 4\hat{i} + 2\hat{j} + 3\hat{k}$

Soln.

$$\text{A unit vector, } \hat{b} = \frac{\vec{b}}{|\vec{b}|}$$

$$|\vec{b}| = \sqrt{4^2 + 2^2 + 3^2} = \sqrt{29}$$

$$\hat{b} = \frac{4i + 2j + 3k}{\sqrt{29}} = \frac{4}{\sqrt{29}}i + \frac{2}{\sqrt{29}}j + \frac{3}{\sqrt{29}}k$$

3. If a force $(2i + j + k)N$ is applied to a load so that it moves a distance $(i - k) m$.

Calculate the work done on the load.

Soln.

$$\begin{aligned} W &= \vec{F} \cdot \vec{S} = (2i + j + k) \cdot (i - k) \\ &= (2 \times 1) + (1 \times 0) + (1 \times -1) = 2 + 0 - 1 = 1 J \\ W &= 1 J \end{aligned}$$

4. A particle at a position $r = i + 2j + 3k$ is acted upon by a force $F_1 = 4i + 5j + 6k$.

Evaluate the torque experienced by the particle.

Soln.

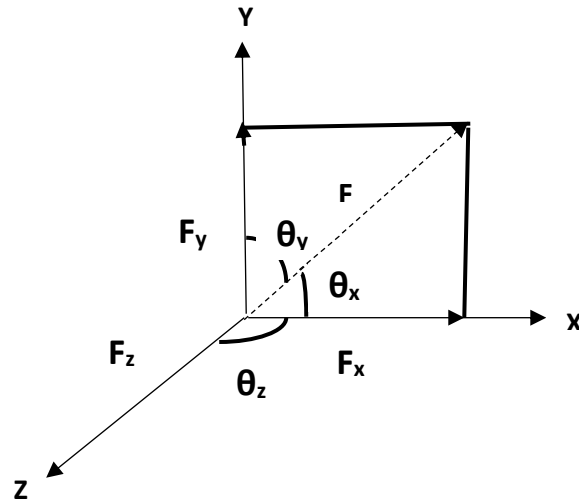
$$\begin{aligned} \text{Torque } \vec{\tau} &= \vec{F} \times \vec{r} = \begin{vmatrix} i & j & k \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix} = i \begin{vmatrix} 5 & 6 \\ 2 & 3 \end{vmatrix} - j \begin{vmatrix} 4 & 6 \\ 1 & 3 \end{vmatrix} + k \begin{vmatrix} 4 & 5 \\ 1 & 2 \end{vmatrix} \\ &= i(15 - 12) - j(12 - 6) + k(8 - 5) \\ \vec{\tau} &= (3i - 6j + 3k)Nm \end{aligned}$$

Hence, the magnitude of torque is; $\tau = |\vec{\tau}| = \sqrt{(3)^2 + (-6)^2 + (3)^2} = \sqrt{54}$

$$\tau = 7.32 Nm$$

Components of a Force

A force acting conjointly with one or more forces at some angle from the coordinate axes can be resolved into mutually perpendicular forces called components. The component of a force parallel to the x-axis is called the x-component, parallel to y-axis is called y-component and parallel to the z-axis is called the z-component.



$$F_x = F \sin \theta_x$$

$$F_y = F \cos \theta_y$$

$$F = \sqrt{F_x^2 + F_y^2}$$

$$\tan \theta_x = \frac{F_y}{F_x}$$

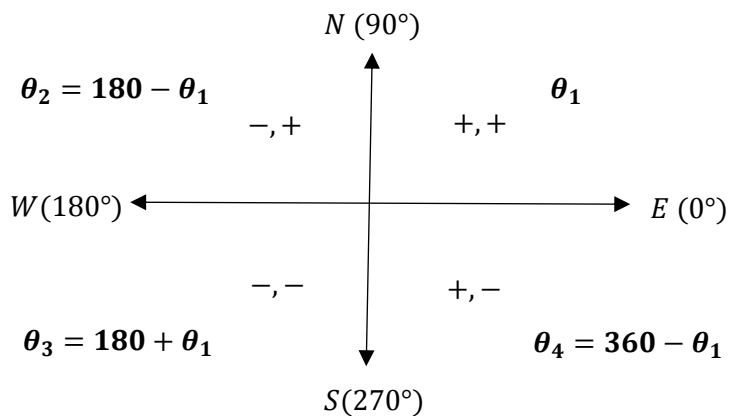
Worked examples

1. Consider a set of forces $f_1 = 100 \text{ N}$ at a Northward direction, $f_2 = 200 \text{ N}$ at angle 20°

North-east direction and $f_3 = 50 \text{ N}$ at a Westward direction. Calculate the following:

- i. The resultant force, F_R and its reference angle θ_R .
- ii. The force at equilibrium, F_E and its reference angle, θ_E .

Soln.

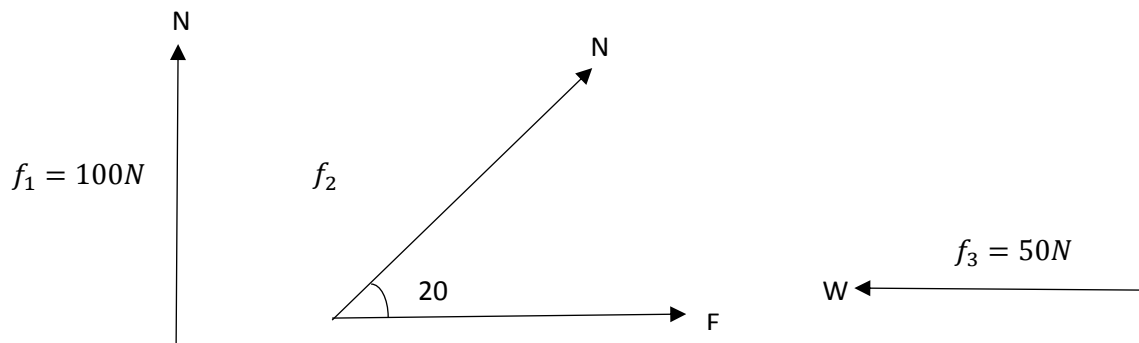


Where $\theta_1 = 1st \text{ Quadrant}$

$\theta_2 = 2nd \text{ Quadrant}$

$\theta_3 = 3rd \text{ Quadrant}$

$\theta_4 = 4th \text{ Quadrant}$



$$F_1X = f_1 \cos \theta = 100 \cos 90^\circ = 0$$

$$F_2X = f_2 \cos \theta = 200 \cos 20^\circ = 187.94$$

$$F_3X = f_3 \cos \theta = 50 \cos 180^\circ = -50$$

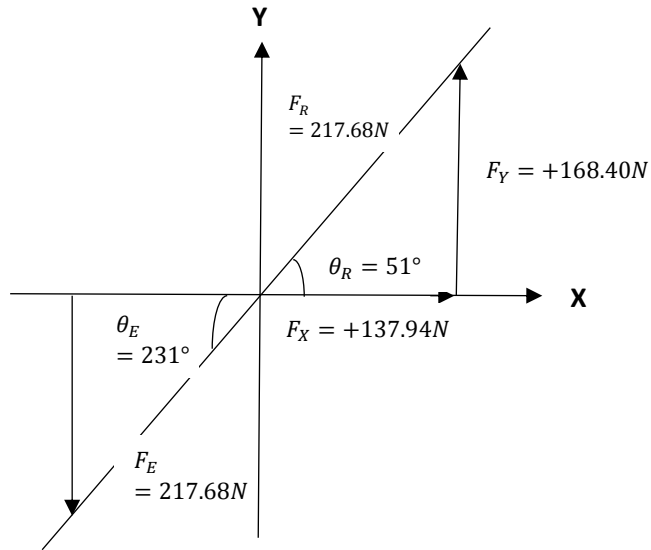
$$F_X = 137.94$$

$$F_1Y = f_1 \sin \theta = 100 \sin 90^\circ = 100$$

$$F_2Y = f_2 \sin \theta = 200 \sin 20^\circ = 68.40$$

$$F_3Y = f_3 \sin \theta = 50 \sin 180^\circ = 0$$

$$F_Y = 168.40$$



i. $F_R = \sqrt{F_X^2 + F_Y^2} = \sqrt{137.94^2 + 168.40^2} = 217.68 \text{ N}$

$\therefore$ The resultant force, $F_R = 217.68 \text{ N}$

$$\theta_R = \tan^{-1} \left(\frac{F_Y}{F_X} \right) = \tan^{-1} \left(\frac{168.40}{137.94} \right) = 50.68^\circ$$

$\therefore$ Its reference angle, $\theta_R = 51^\circ$

ii. $F_E = \sqrt{F_X^2 + F_Y^2} = \sqrt{(-137.94)^2 + (-168.40)^2} = 217.68 \text{ N}$

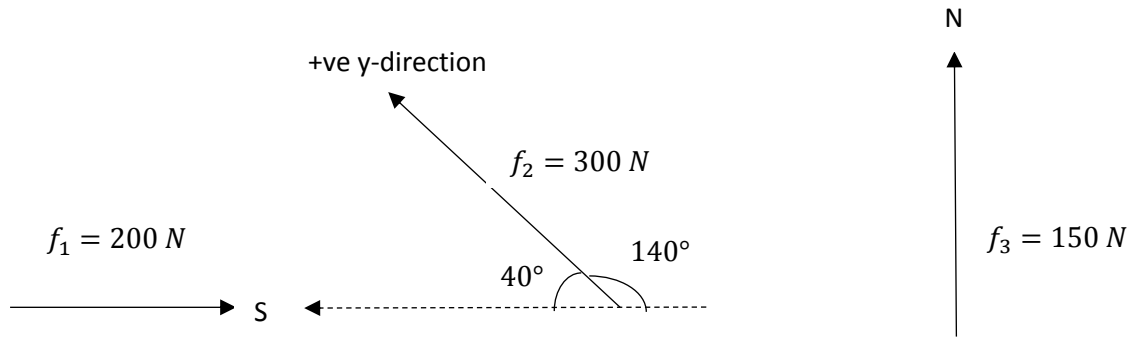
$\therefore$ The force at equilibrium, $F_E = 217.68 \text{ N}$

Also, the net force $F_{Net} = F_E - F_R = 217.68 - 217.68 = 0$. Note that $F_R = F_E$

$$\theta_E = 180^\circ + \theta_R = 180^\circ + 51^\circ = 230.68^\circ$$

$\therefore$ Its reference angle, $\theta_E = 231^\circ$

2.



- i. Find the resultant force vector, F_R of the above three (3) vectors.
- ii. Find the reference angle for the resultant force vector and show its angle along the +ve x-direction.
- iii. Find the force at equilibrium, F_E
- iv. Find the angle for the force at equilibrium.

Soln.

$$F_1X = f_1 \cos \theta = 200 \cos 0^\circ = 200$$

$$F_2X = -f_2 \cos \theta = -300 \cos 40^\circ = -229.81$$

or

$$= f_2 \cos(180 - \theta) = 300 \cos 140^\circ = -229.81$$

$$F_3X = f_3 \cos \theta = 150 \cos 90^\circ = 0$$

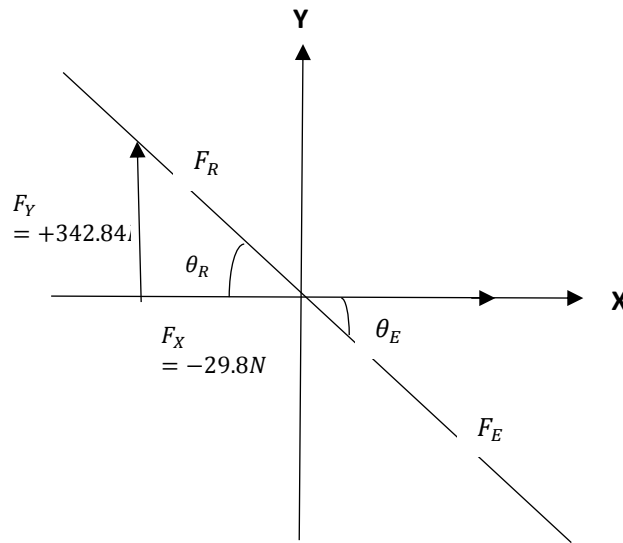
$$F_X = -29.81$$

$$F_1Y = f_1 \sin \theta = 200 \sin 0^\circ = 0$$

$$F_2Y = f_2 \sin \theta = +300 \sin 40^\circ = 192.84$$

$$F_3Y = f_3 \sin \theta = 150 \sin 90^\circ = 150$$

$$F_Y = 342.84$$



i. The resultant force, $F_R = \sqrt{F_X^2 + F_Y^2} = \sqrt{(-29.81)^2 + (342.84)^2} = 344.13 \text{ N}$

$\therefore$ The resultant force, $F_R = 344.13 \text{ N}$

$$\theta_R = \tan^{-1}\left(\frac{F_Y}{F_X}\right) = \tan^{-1}\left(\frac{342.84}{-29.81}\right) = -85.05^\circ$$

i.e. the -ve sign shows that the ref. angle is at -ve direction.

$\therefore$ Its reference angle, $\theta_R = 180 - 85.05^\circ = 94.95^\circ$

ii. $F_E = \sqrt{F_X^2 + F_Y^2} = \sqrt{(+29.81)^2 + (-342.84)^2} = 344.13 \text{ N}$

$\therefore$ The force at equilibrium, $F_E = 344.13 \text{ N}$

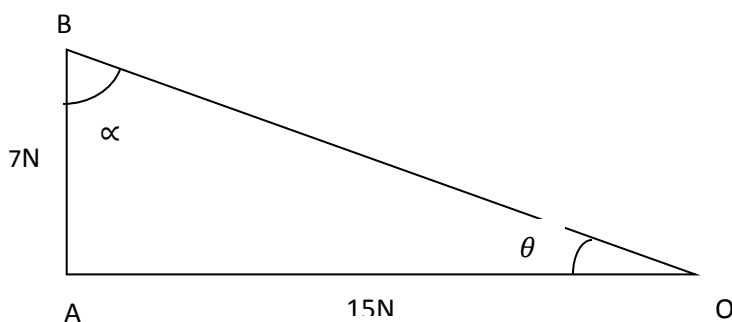
Also, the net force $F_{Net} = F_E - F_R = 344.13 - 344.13 = 0$. Note that $F_R = F_E$

$$\theta_E = 360^\circ - \theta_R = 360^\circ - 94.95^\circ = 265.05^\circ$$

$\therefore$ Its reference angle, $\theta_E = 265.05^\circ$

EXERCISE (SCALAR & VECTORS)

1. Find by the means of vector diagram, the direction/angles and the resultant of 2 forces of 7N and 15N acting perpendicular to each other.

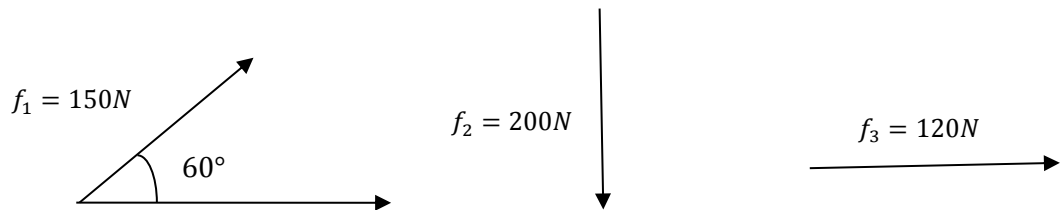


2. Define the angular momentum of a rotating system through a radius, $\vec{r}$. hence, calculate the angular momentum $\vec{L}$ of the system of mass 3kg moving with a velocity $\vec{V} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ through a circular path of radius $\vec{r} = -3\mathbf{i} + \mathbf{j} + 3\mathbf{k}$.
3. Consider the displacement vectors $\vec{A} = (3\mathbf{i} + 3\mathbf{j})\text{ m}$, $\vec{B} = (\mathbf{i} - 4\mathbf{j})\text{ m}$, and $\vec{C} = (-2\mathbf{i} + 5\mathbf{j})\text{ m}$. Use the component method to determine;
 - (a) the magnitude and direction of the vector $\vec{D} = \vec{A} + \vec{B} + \vec{C}$
 - (b) the magnitude and direction of $\vec{E} = -\vec{A} - \vec{B} + \vec{C}$.
4. Consider two vectors $\vec{A} = 3\mathbf{i} + 2\mathbf{j}$ and $\vec{B} = -\mathbf{i} - \mathbf{j}$, calculate:
 - i. $A + B$
 - ii. $A - B$
 - iii. $|A + B|$
 - iv. $|A - B|$
 - v. The direction of $A + B$ and $A - B$.

5. Given three (3) vectors $f_1 = 300\text{N}$ south, $f_2 = 400\text{N}$ west and $f_3 = 500\text{N}$ at an angle 50° north east. Calculate the following:

- The magnitude of the resultant vector and its reference angle.
- The equilibrium force and its angle.

6.



Calculate:

- The magnitude of the resultant force vector and its reference angle at +ve x-direction.
 - The force at equilibrium and its angle.
 - The net force
7. Let P and Q be two (2) forces with magnitude 6N and 12N respectively and the angle between them be θ . Let R be the resultant force and ϕ be the angle between P and R. Calculate:

- The resultant force
- The angle ϕ

8. Two forces of magnitude 3N and 2N are inclined at 180° to each other act on the body. Find the resultant of the two (2) forces and the angle between them.

9. If $a = \langle 5, -4, 3 \rangle$ and $b = \langle -7, 2, -8 \rangle$, calculate;

i. $\vec{a} \times \vec{b}$

ii. $\vec{a} \cdot \vec{b}$

10. Given that $P = -4i + 7j + k$ and $Q = 3i - 4j + 2k$

i. Find the dot product of P and Q

ii. What is the angle between P and Q

iii. Find the projection of P on Q

iv. Find the projection of Q on P

11. A force $F = 2i + 7j - 2k$ acts on a particle, producing a displacement of $x = 3i + j -$

$2k$. What is the work-done on the particle?

12. Calculate the magnitude and direction of the torque due to a force $F = (2i - 3j +$

$4k)$ N acting at a point $r = 3i + 2j + 3k$ about the origin.

13. If $A = 2i - j + 2k$, $B = 5i + 6j - 2k$; Find (i) $|A|$ (ii) $|B|$ (iii) $|A \cdot B|$ (iv) the angle between A and B .